

Signsprout business plan

Prepared by Shivam Gupta, November 2026. Figures marked "estimate" are assumptions to be validated; everything else has a source.

1. Summary

Signsprout teaches first ASL vocabulary on Meta Quest with the learner's own hands: Sprout, a friendly practice buddy, demonstrates each sign face to face, guide hands overlay the learner's hands, and the app reads 25 joints per hand, checks each finger and says what to fix. It starts with the people who need it most, hearing parents of deaf babies, and grows into a practice tool for anyone learning ASL.

- **Product:** a WebXR app (a link, no install) plus a phone companion. Built and tested: 87 signs, spaced repetition, the step-by-step checker and the companion site (accounts, pairing, sync and the practice coach run on Supabase; the public preview works fully offline). Not launched yet, with no users, pilots or revenue so far.
 - **Model (planned):** free core; a Family plan at **\$7.99/month or \$59/year**; sponsored free access for families of deaf children under three, funded through partner programs; program licences for early-intervention services and schools (from \$39 per family per year).
 - **Costs:** static hosting and a managed database; under **\$0.05 per active family per month** at the scale of the first year (estimate, \$6).
 - **Ask:** Deaf partners to co-lead content (paid, with a veto over what we teach), three pilot programs, and a Meta VR Store listing in Q1 2027.
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2. The problem

- More than **90% of deaf children are born to hearing parents** (NIDCD).
- Deaf children of hearing parents reach age-level ASL vocabulary when exposed **before 6 months**; later exposure leads to smaller vocabularies (Caselli et al. 2021).
- Only **22.9%** of families with deaf children regularly sign at home (Lieberman et al. 2024, citing GRI 2013–14 data). Parents cite **time** and **cost**; rural families cite access.
- Learning from video is hard for a specific reason: hearing non-signers copying a face-to-face model made errors on **24.3%** of sideways (lateral) movements; in a second study, a

model that matched their mirroring cut lateral errors from **20.6% to 0.9%** (Shield & Meier 2018). And video can't tell you if your hands are right.

3. The product

Capability	Why it matters
Sprout demonstrates face to face, mirrored by default	Removes the mental flip behind many novice errors
Guide hands on the learner's own hands	First-person demonstration; fades after two unaided successes to avoid dependence
A checker that goes in order: handshape, then place, then movement, then the final handshape	"Fold your ring finger down more" instead of pass/fail
A mirror view beside Sprout	Shows your hands against your face for signs made at the face
Five-minute sessions, spaced repetition, a garden	A habit that fits a parent's day; visible progress
practice coach that plans only from signs Signsprout can check	Personal ("bath time words") without inventing signs
Accessibility built in	Captions for every prompt, optional voice read-out, left-handed mode, adjustable strictness and guide speed, limited finger range, high contrast, calm motion, a one-hand option
Phone pairing, family sharing	No typing in VR; grandparents and carers can follow along
Web first	A link, no store approval, updates instantly, runs on Quest 2, 3, 3S and future VR glasses

4. Market

Beachhead: hearing families of deaf and hard-of-hearing (DHH) children. About **6,272** US infants were identified as DHH in 2022 (CDC EHDI), so roughly **5,600** new hearing-parent families a year and about **28,000** with a child under five. Each family has several learners (parents, siblings, grandparents, carers). This segment is small as a consumer revenue pool but large in need, credibility and program funding: IDEA Part C early intervention receives **\$540M a year** federally (CRS), and every state's EHDI program has language acquisition as a goal (GAO-25-106978).

Expansion: general ASL learners. - ASL is the **#3 language** in US higher education, with **107,899** enrollments, and one of only three top-15 languages that grew from 2016 to 2021 (MLA). - **2.8%** of US adults use sign language (Mitchell & Young 2023). - Lingvano grew from 500k to **2.5M learners** (2022–2024) with 2D video (eSchool News). Duolingo still has no ASL course.

Platform. Meta held 74.6% of AR/VR headset share in 2024, and education was its fastest-growing segment (+69.4%) (IDC). Meta's new VR glasses are hands-first by design, and hand-tracked titles are shown first to hands-only users. The main risk is headset reach (see §8), which is why Signsprout is a web app with a phone companion and a lending model.

5. Business model

Plan	Price	Who
Free	\$0	Everyone: the First words, Mealtime and Family units, fingerspelling A–Z and numbers, the garden
Family	\$7.99/month or \$59/year, up to 4 family members	All signs and new units, practice coach, sync, family dashboard and share links
Family Pass (sponsored, planned)	\$0 to the family	Families with a DHH child under 3, referred by early intervention, EHDI or a hospital, funded by programs and philanthropy
Programs	from \$39 per family per year	Early-intervention services, audiology teams, schools for the deaf: seats, consented progress reports, headset lending kits, staff and Deaf-mentor onboarding

All prices are planned launch pricing; the preview is free. Pricing sits below Lingvano (\$119.99/year) and ASL Bloom (\$98.99/year), and near Duolingo Family (\$119.99/year for six) per learner. **The plan is that families of deaf babies never pay**, funded through partner programs: that is both the mission and the channel, because program staff recommend tools they can hand out for free.

Non-dilutive funding fits this product: NSF SBIR (Phase I up to \$305,000), ED/IES SBIR (\$250,000) and NIDILRR, all reauthorised through 2031.

6. Costs

The product is a static web app plus a small database, so costs stay low and mostly fixed.

Item	Free tier	At about 10,000 monthly families (estimate)
Hosting (static, CDN)	\$0	\$0–20/month
Supabase (auth, Postgres, functions)	\$0 (free plan)	\$25/month (Pro) plus usage
practice coach (Gemini)	Rules planner, \$0	A few cents per plan, capped at 20 plans per user per day; most families make one or two a week
Email (sign-in links)	Supabase built-in	About \$15/month
Total	\$0	Roughly \$100–500/month , under \$0.05 per family

Hand tracking runs on the headset, so there are no per-user inference costs for the core loop. Content is data (a sign is a few lines), so adding signs costs review time rather than video production.

7. Competition and moat

	Signsprout	NVIDIA Signs	Lingvano	PopSign	Quest ASL apps
Checks your own hands	✓ 3D, 25 joints/hand	✓ 2D webcam	Self-view camera	✓ phone camera	✓ (mostly letters)
Guide hands on yours (first person)	✓	✗	✗	✗	✗
Explains what to fix	✓ per finger and parameter	Real-time feedback (2D)	✗	✗	Limited
Family curriculum + spaced repetition	✓	✗	✓	✓ (game)	✗
Price	Free core	Free	\$119.99/yr	Free	Varies

Moat. (1) The sign engine: signs written as linguistic parameters, so one definition drives Sprout, the guide hands, instructions and checking, with no training data needed per sign. Adding a sign takes minutes, and new sign languages reuse the engine. (2) Trust, which we still have to earn: paid review of every sign by Deaf signers before launch, Deaf co-leadership of content with pay and a veto (the plan, not yet in place), and a privacy stance (hand data never leaves the headset). (3) Distribution through early-intervention programs, which consumer apps rarely reach. (4) Progress data (which signs are hard, where) improves hints and the curriculum.

8. Risks and mitigations

Risk	Mitigation
Few families own a headset	Web app plus phone companion; program lending kits; the same engine can drive a phone-camera mode
Checker accuracy on real hands is not yet measured	Tests so far use a simulated learner (synthetic hands built from the same hand model, not yet real-user data); on-device testing with real signers is next
Hand-tracking limits (occlusion, similar handshapes)	For fingerspelled letters that hand tracking confuses (such as M, N, T, E, A, S), Signsprout is lenient and says so; specific hints; skip always available; no over-claiming
Community trust	No Deaf signer has reviewed Signsprout yet: paid review by Deaf signers comes before launch, and Deaf co-leadership (paid, with a veto over content) is the plan; a practice partner, never a translator; the site links to Deaf-led resources and Deaf Mentor programs
Headset market softness (IDC: VR headsets fell in 2025)	Web-first, runs on current and upcoming Meta devices, not tied to one store

9. Roadmap

When	Milestone
Nov 2026	Competition build: 87 signs, coach, pairing, companion site
Dec 2026	Open web beta; on-device testing with real signers; recruit paid Deaf reviewers and three pilot programs
Q1 2027	Paid review of every sign by Deaf signers, then the Meta VR Store listing and Family plan
Q2 2027	Facial-grammar lessons (taught with video, not scored); two-person practice
H2 2027	Phone-camera mode; BSL and ISL with local Deaf partners

10. Team

Shivam Gupta, creator: product, design and engineering. Looking for: a Deaf co-founder or content lead, advisors from early intervention and Deaf education, and pilot partners.